

KDM5A fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes KDM5A gene fusions from the msk_impact_50k_2026 study, covering 10 fusion events. 3/6 fusions with known frame status (50.0%) were in-frame. 5/9 fusions (55.6%) retained the PLU-1-like protein domain. Domain retention was tested with Fisher's exact test ($p=0.119048$, not statistically significant at $\alpha=0.05$). No literature benchmark is configured for KDM5A.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 10 analyzed fusion events, identifying 9 distinct fusion partner genes. The most frequent partner was NINJ2, observed in 2 events.

Domain retention

PLU-1-like protein domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 3/3 (100.0%) of in-frame fusions retained the domain, $p=0.119048$ (not statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.237624.

Domain disruption

Disruption of the jmjN domain, ARID/BRIGHT DNA binding domain, PHD-finger, JmjC domain, hydroxylase, and C5HC2 zinc finger was tested with Fisher's exact test comparing in-frame fusions against all others; 1/3 (33.3%) of in-frame fusions disrupted the domain, $p=0.761905$ (not statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.821782.

Cutpoint detection

Cutpoint detection scanned 9 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 1311 aa (exon 24; permutation-corrected $p=0.267327$, not statistically significant at $\alpha=0.05$). This is 94 aa from the nearest configured domain boundary at 1217 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" ($n=5$) and "not_retained" ($n=4$). For Frame_status, retained: 3/4 (75.0%, 95% CI 30.1%-95.4%); not_retained: 0/2 (0.0%, 95% CI 0.0%-65.8%). Welch's t-test comparing Tumor_variant_count between groups gave means of 15.6 vs 17.25, $p=0.843016$ (not statistically significant at $\alpha=0.05$).

Composite evidence score

Composite evidence score produced results for this run (9 row(s) in composite_evidence_ranking); see the accompanying table.

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Expression association

Expression association reported: No mRNA expression data was available for KDM5A in this cohort; expression-association analysis was skipped.

Genomic position recurrence

Genomic position recurrence produced results for this run (8 row(s) in genomic_bin_recurrence, 0 row(s) in exact_position_recurrence, 0 row(s) in protein_position_genomic_spread); see the accompanying table.

Joint partner

Joint partner reported: KDM5A has no gene_pair configured; joint-partner analysis was skipped.

Mechanistic interpretation

Mechanistic interpretation produced results for this run; see the accompanying data for details.

Mutation cooccurrence

Mutation cooccurrence reported: KDM5A has no mutual_exclusivity_targets configured; co-occurrence/mutual-exclusivity analysis was skipped.

Window detection

Window detection reported: no candidate window of the configured widths (25, 50, 100, 200) met the minimum-event-count guard (min_events_per_window=4) on both sides of the split

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
COLGALT1	1	0.1	0.06241101320710366	0.00852432814065687	1.0	0.34430466544937777	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.23173808547484506	1
DIPK1A	1	0.1	0.06241101320710366	0.00852432814065687	0.9581105169340464	0.34430466544937777	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.225454663014952	2
DHX37	1	0.1	0.06241101320710366	0.00852432814065687	0.8449197860962567	0.34430466544937777	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.20847605338928357	3
NINJ2	2	0.2	0.06241101320710366	0.00852432814065687	0.6154188948306596	0.34430466544937777	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.204050919699444	4
BMAL2	1	0.1	0.06241101320710366	0.00852432814065687	0.4233511586452763	0.34430466544937777	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.1452407592716365	5
USHBP1	1	0.1	0.06241101320710366	0.00852432814065687	None	0.34430466544937777	confidence_certainty, domain_disruption, domain_retention, recurrence	0.09616245349981772	6
CACNA1C	1	0.1	0.06241101320710366	0.00852432814065687	0.03208556149732622	0.34430466544937777	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.086550919699444	7
STAT5A	1	0.1	0.06241101320710366	0.00852432814065687	0.031194295900178304	0.34430466544937777	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.08641722985987181	8
OSBPL8	1	0.1	0.06241101320710366	0.00852432814065687	0.0	0.34430466544937777	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.08173808547484507	9

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
189	0	5	1	3	0.0	0.4444444444444445	0.3521825181113625
224	0	5	2	2	0.0	0.16666666666666669	0.7781512503836436
225	1	4	2	2	0.25	0.523809523809524	0.2808266095756941
664	2	3	2	2	0.6666666666666666	1.0	-0.0
759	2	3	3	1	0.2222222222222222	0.5238095238095237	0.2808266095756943
1311	2	3	4	0	0.0	0.16666666666666663	0.7781512503836437
1358	3	2	4	0	0.0	0.4444444444444444	0.35218251811136253
1485	4	1	4	0	0.0	1.0	-0.0

domain_disruption tables

frame_domain_contingency_table

1	2
2	4

domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_disruption_pf02375	jmjN domain	9	6	0	3	0.0
auto_disruption_pf01388	ARID/BRIGHT DNA binding domain	9	6	0	3	0.0
auto_disruption_pf00628	PHD-finger	9	5	0	4	0.0
auto_disruption_pf02373	JmjC domain, hydroxylase	9	5	0	4	0.0
auto_disruption_pf02928	C5HC2 zinc finger	9	6	0	3	0.0

domain_retention tables

frame_domain_contingency_table

3	2
0	4

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_key_domain	PLU-1-like protein	9	5	1	3	0.015015015015015015

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
16	6	10	0.6

frequency tables

Partner_gene_counts

Partner_gene	Event_count
BMAL2	1
CACNA1C	1
COLGALT1	1
DHX37	1
DIPK1A	1
NINJ2	2
OSBPL8	1
STAT5A	1
USHBP1	1

genomic_position_recurrence tables

genomic_bin_recurrence

chromosome	bin_start	bin_end	n_events	event_ids
12	432000	433000	2	EVT-P-0030430-T03-IM7-42, EVT-P-0030719-T01-IM6-4
12	401000	402000	1	EVT-P-0060766-T02-IM7-41
12	402000	403000	1	EVT-P-0064343-T01-IM7-8
12	415000	416000	1	EVT-P-0065027-T03-IM7-9
12	416000	417000	1	EVT-P-0021409-T01-IM6-1
12	465000	466000	1	EVT-P-0030473-T01-IM6-5
12	470000	471000	1	EVT-P-0039045-T01-IM6-44

chromosome	bin_start	bin_end	n_events	event_ids
12	472000	473000	1	EVT-P-0022798-T01-IM6-43

Per-event results (results.tsv)

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frameshift	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0021409-T01-IM6-1	P-0021409-T01-IM6		Ovarian Cancer	HGSOC	COLGALT1-KDM5A	COLGALT1	unknown	three_prime	416253	24	1311	False	1311	True	exact match	lost	0.0	False		PLU-1-like protein; jmjN domain; ARID/BRIGT DNA binding domain; PHD-finger; JmjC domain, hydroxylase; C5HC2 zinc finger		COLGALT1 (NM_024656)-KDM5A (NM_001042603) rearrangement: t(19;12)(p13.11;p13.33)(chr19:g.17669210::chr12:g.416253) Protein Fusion: mid-exon {COLGALT1:KDM5A}	NA
EVT-P-0030719-T01-IM6-4	P-0030719-T01-IM6		Non-Small Cell Lung Cancer	LUAD	KDM5A-BMAL2	BMAL2	unknown	three_prime	432924	15	664	False	664	True	exact match	retained	1.0	False	PLU-1-like protein; PHD-finger; C5HC2 zinc finger	jmjN domain; ARID/BRIGT DNA binding domain; JmjC domain, hydroxylase		ARNTL2 (NM_001248002) - KDM5A (NM_001042603) rearrangement: c.285-4357:ARNTL2_c.1992:KDM5A Protein Fusion: mid-exon {ARNTL2:KDM5A}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0030473-T01-IM6-5	P-0030473-T01-IM6		Breast Cancer	IDC	KDM5A-CACNA1C	CACNA1C	in-frame	three_prime	465876	6	225	True	225	True	exact match	retained	1.0	False	PLU-1-like protein; PHD-finger; JmjC domain, hydroxylase; C5HC2 zinc finger	jmjN domain; ARID/BRIGT DNA binding domain		CACNA1C (NM_001129827) - KDM5A (NM_001042603) rearrangement: c.1114-2453: CACNA1C_c.673-173: KDM5A_in frame {CACNA1C:KDM5A}	NA
EVT-P-0064343-T01-IM7-8	P-0064343-T01-IM7		Soft Tissue Sarcoma	MFH	KDM5A-DHX37	DHX37	out-of-frame	five_prime	402339	26	1485	True	1485	True	exact match	retained	1.0	False	PLU-1-like protein; jmjN domain; ARID/BRIGT DNA binding domain; PHD-finger; JmjC domain, hydroxylase; C5HC2 zinc finger			KDM5A (NM_001042603) - DHX37 (NM_001032656) rearrangement: c.4456-4: KDM5A_c.2045+567: DHX37dup Protein Fusion: out of frame {KDM5A:DHX37}	NA

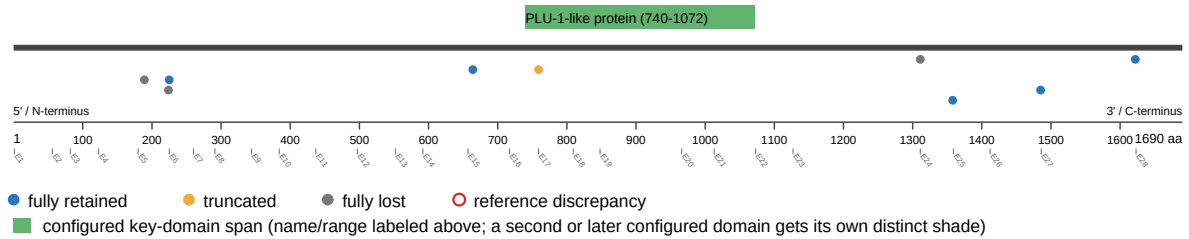
event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0065027-T03-IM7-9	P-0065027-T03-IM7		Esophagogastric Cancer	GEJ	KDM5A-DIPK1A	DIPK1A	in-frame	five_prime	415999	24	1358	True	1358	True	exact match	retained	1.0	False	PLU-1-like protein; JmjN domain; ARID/BRIGT DNA binding domain; PHD-finger; JmjC domain, hydroxylase; C5HC2 zinc finger			KDM5A (NM_001042603) - FAM69A (NM_001006605) rearrangement: t(1;12)(p22.1;p13.33)(chr1:g.93411462::chr12:g.415999) Protein Fusion: in frame {KDM5A: FAM69A}	NA
EVT-P-0060766-T02-IM7-41	P-0060766-T02-IM7		Soft Tissue Sarcoma	LMS	KDM5A-NINJ2	NINJ2	in-frame	five_prime	401803	27	1622	True	1622	True	exact match	retained	1.0	False	PLU-1-like protein; JmjN domain; ARID/BRIGT DNA binding domain; PHD-finger; JmjC domain, hydroxylase; C5HC2 zinc finger			KDM5A (NM_001042603) - NINJ2 (NM_016533) rearrangement: c.4866+122:KDM5A_c.172-28503:NINJ2dup Protein Fusion: in frame {KDM5A: NINJ2}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0030430-T03-IM7-42	P-0030430-T03-IM7		Non-Small Cell Lung Cancer	LUAD	KDM5A-NINJ2	NINJ2	out-of-frame	five_prime	432137	16	759	True	758	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.06006006006006	True	jmjN domain; ARID/BRIGHT DNA binding domain; JmjC domain, hydroxylase; C5HC2 zinc finger	PHD-finger	PLU-1-like protein	KDM5A (NM_001042603) - NINJ2 (NM_016533) rearrangement: c.2275+111:KDM5A_c.172-20595:NINJ2dup Protein Fusion: out of frame {KDM5A:NINJ2}	NA
EVT-P-0022798-T01-IM6-43	P-0022798-T01-IM6		Prostate Cancer	PRAD	KDM5A-OSBPL8	OSBPL8	unknown	five_prime	472234	5	189	False	189	True	exact match	lost	0.0	False	jmjN domain; ARID/BRIGHT DNA binding domain	PLU-1-like protein; PHD-finger; JmjC domain, hydroxylase; C5HC2 zinc finger		KDM5A (NM_001042603) - OSBPL8 (NM_020841) Rearrangement: c.567:KDM5A_c.1630+2182:OSBPL8dup Protein Fusion: mid-exon {KDM5A:OSBPL8}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frameshift_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0039045-T01-IM6-44	P-0039045-T01-IM6		Ovarian Cancer	CCOV	KDM5A-STAT5A	STAT5A	out-of-frame	five_prime	470840	5	224	True	224	True	exact match	lost	0.0	False	jmjN domain; ARID/BRIGHT DNA binding domain	PLU-1-like protein; PHD-finger; JmjC domain, hydroxylase; C5HC2 zinc finger		KDM5A (NM_001042603) - STAT5A (NM_003152) rearrangement: t(12;17)(p13.3;q21.2)(chr12:g.470840::chr17:g.40459887) Protein Fusion: out of frame {KDM5A:STAT5A}	NA

domain retention outliers

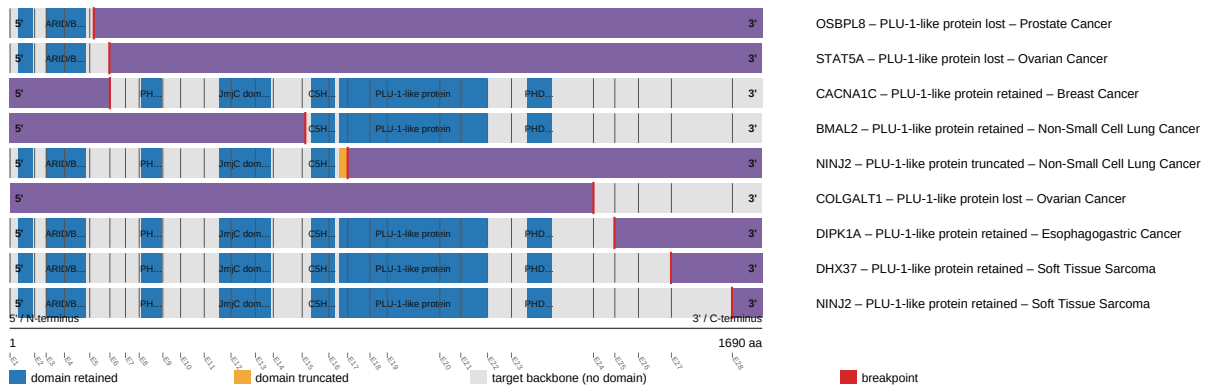
KDM5A domain-retention track



fusion schematic

KDM5A fusion-transcript schematic

Fusion order varies by row; follow each row's 5' and 3' labels (KDM5A portion is domain-colored, exon ticks). Partner-gene blocks share one color; row labels identify each partner.



Partner-block colors are decorative; row labels identify partners.

intragenic deletion schematic

KDM5A intragenic-deletion schematic

retained N-terminal block → deleted span (plain connector) → resumed C-terminal block

